APPLICATION OF ADAPTIVE FILTERS IN ACTIVE POWER FILTERS

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Abstract – **This paper presents a new idea applied to adaptive filters for harmonic detection in shunt active power filters. Improve the speed of convergence of the adaptive filter and reduces the steady-state error are the finals objectives of the strategy. Two cases are presented and discussed, one with a Finite Impulse Response (FIR) adaptive filter with 32 coefficients and another with an adaptive notch filter. The Least Mean Square (LMS) algorithm was used to adjust the coefficients in the both cases. An algorithm based on Coulon oscillator principle was used to determine the load current amplitude and then modify the step-size value. Simulations using Matlab are presented to clarify the algorithm. Also practical implementation is performed using the DSP Texas Instruments TMS320F2812 and the experimental results depicted.**

Keywords – **Adaptive filters, active power filter, adaptive FIR filter, adaptive notch filter, harmonic compensation.**

I. INTRODUCTION

The compensation for harmonic currents, generated by power electronic equipments, has become important as its growing use in the industrial applications. In this context, more and more, new techniques are developed in order to improve the effectiveness of active power filters (APF) in harmonic mitigation. The performance of active power filters basically depend on the extraction method used to generate the current reference signal, the control method used to generate the compensation harmonic current and the dynamic characteristic of the entire filter [1].

In power system, the current can be described as

$$i(t) = \sum_{n=1}^{N} A_n \sin(n\omega t + \theta_n) \quad (1)$$

The fundamental current is denoted by $i_1(t)$ and harmonic current as

$$i_h(t) = \sum_{n=2}^{N} A_n \sin(n\omega t + \theta_n) \quad (2)$$

The harmonics can be obtained just eliminating the fundamental components. The problem arisen is how to extract the sinusoidal waveform, separating the harmonic components.

Actually the synchronous reference frame and the instantaneous power theory are the two methods of harmonic extraction more used in active power filters control. In recent years the use of Adaptive Filters has shown a powerful technique to perform this job.

II. HARMONIC DETECTION WITH ADAPTIVE FILTERS

The adaptive noise canceling technique based on the Wiener theory has been widely used in many signal processing applications [2]. It can maintain the system in the best operating state by continuously self-adjusting its parameters. According to the adaptive noise canceling theory, the principle of adaptive harmonic detecting method is that illustrated in Fig 1.

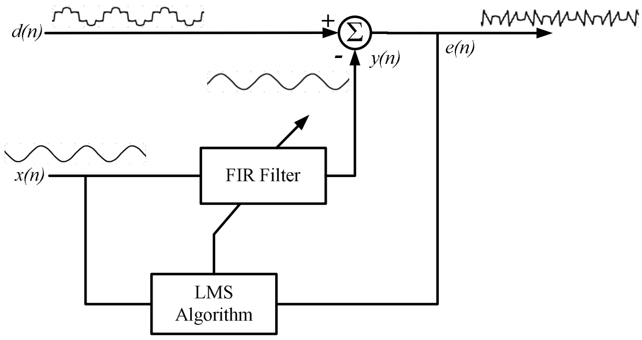

Fig. 1. The principle of adaptive detecting of harmonics.

In this basic structure of adaptive filter (Fig. 1) applied to current harmonic detection, $d(n)$ represents the current $i(t)$ blurred with harmonics and $x(n)$ represents the sinusoidal waveform $v_{sin}(t)$ in phase with the source voltage. The reference input signal $x(n)$ is processed by adaptive filter producing the output signal $y(n)$ that tracks the variation of fundamental signal of the load current. The objective of adaptive filter is approximates $y(n)$, in both amplitude and phase, to the fundamental signal $i_1(n)$. So, the desired harmonics content $i_h(n)$ can be directly obtained from the error signal $e(n)$ given by the subtracting of $y(n)$ from $d(n)$. The coefficients of the adaptive filter are adjusted using the Least Mean Square (LMS) algorithm. The recursion formula of LMS algorithm is given by the equations:

$$e(n) = d(n) - y(n) = d(n) - \boldsymbol{X}^T(n)\boldsymbol{W}(n) \quad (3)$$

$$\boldsymbol{W}(n+1) = \boldsymbol{W}(n) + \mu e(n)\boldsymbol{X}(n) \quad (4)$$

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Where:
$\boldsymbol{X}(n)$ - input vector.
$\boldsymbol{W}(n)$ - coefficient vector.
μ - step-size.

The parameter μ controls the rate of convergence of the algorithm to the optimum solution.

In this work a Finite Impulse Response (FIR) adaptive filter [2-4] with 32 coefficients and an adaptive notch filter [2][5-7] were tested. The adaptive FIR filter structure uses the general structure of adaptive filter shown in Fig.1. The adaptive notch filter is shown in Fig. 2. This structure (Fig. 2) uses two orthogonal signals as input, one for the input $x(n)$ and other 90° phase shifted. In this way only two coefficients are needed to be adapted. The adaptation procedure is the same as used in the general structure of adaptive filters.

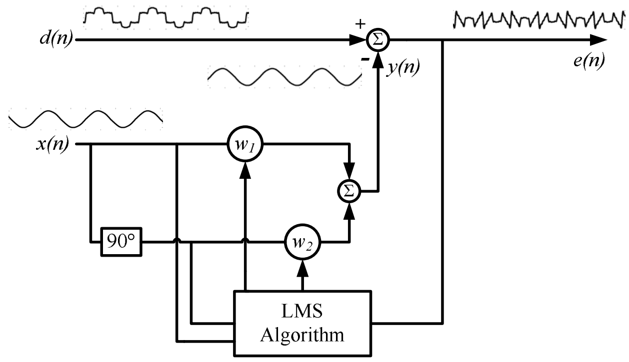

Fig. 2. Adaptive notch filter principle.

To overcome the problems with the speed of convergence, several approaches based on a variable step-size method were proposed in recent years [5-8]. The approach proposed in this work also plays with the step-size to speed-up the adaptation algorithm. But, instead of looking to the error signal, which is polluted with noise complicating the adaptation algorithm behavior, it looks directly to the load current value searching for significative transients. Once the transient in the load current is detected the algorithm starts the step-size adjustments. That shows up to improve the adaptation algorithm behavior. In the simulations the convergence speed reaches the maximum of 1.5 cycles for both cases, the adaptive FIR filter and the adaptive notch filter, against the minimum of 2 cycles reported in the literature [4][6-7]. The amplitude of the load current is determined with a simple algorithm described in [9], based on Coulon oscillator principle. This algorithm is illustrated in Fig. 3.

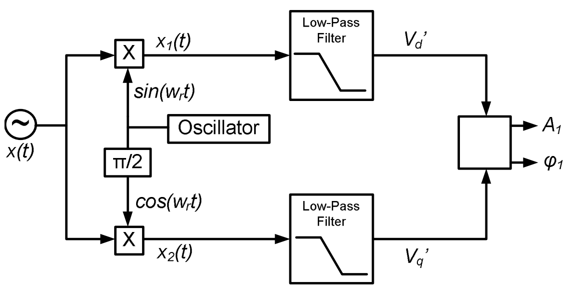

Fig. 3. Coulon oscillator principle.

The oscillator frequency w_r is defined in 60Hz and the load current is used as the signal $x(t)$. In this way, the amplitude of the load current fundamental component can be detected with the algorithm.

With an IIR low-pass filter the components V_d' and V_q'can be extracted as shown in (5) and (6):

$$V_d' = \frac{A_1}{2}\cos(\varphi_1) \tag{5}$$

$$V_q' = \frac{A_1}{2}\sin(\varphi_1) \tag{6}$$

Various filters configurations were designed and tested. A second order elliptic filter with a 30Hz cutoff frequency was chose. With this filter was obtained a fast response of the algorithm presented in the Fig. 3. The frequency response of this filter is presented in the Fig. 4.

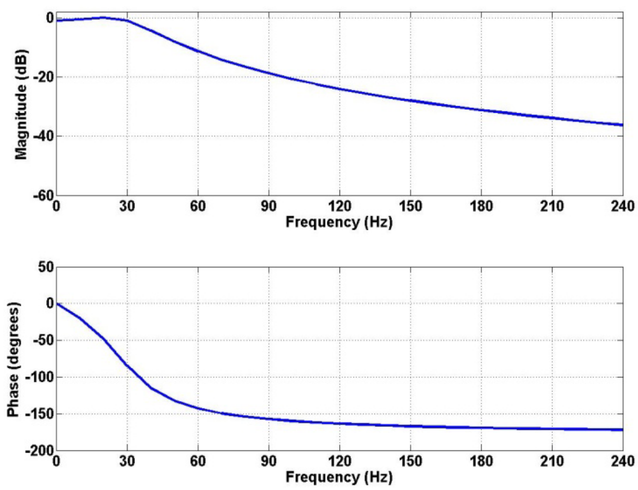

Fig. 4: Frequency response of second order elliptic filter with cutoff frequency of 30Hz.

The amplitude of the load current fundamental component is calculated as demonstrated in (7):

$$A_1 = \sqrt{(2V_d')^2 + (2V_q')^2} \tag{7}$$

When any significant changes happen in the load current value, it is detected and the step-size is immediately increased to a certain value (μ_{max}). After 8.45 ms the step-size is decreased to a medium value (μ_{med}). Finally, after 16.7 ms, μ is modified to a minimum or steady-state value (μ_{min}) and stay with this value until next transient in the load current. The values of μ are defined as

$$\mu_{min} < \mu_{med} < \mu_{max} \tag{8}$$

This procedure guaranties a fast transient response to adaptive filter action.

III. SIMULATION RESULTS

The simulation results are shown in Figs. 5, 6, 7 and 8. The transient behavior of the adaptive FIR filter strategy, for a 100% load current variation, is presented in Fig. 5 and Fig. 6. The Fig. 5 shows the signals: the load current $d(n)$ in the first graphic, the fundamental of the load current (in blue) and the adaptive FIR filter output $y(n)$ (in red) in the second graphic and the harmonic current reference $e(n)$ in the third graphic. The Fig. 6 shows the desired harmonic reference and the harmonic reference generated by the adaptive FIR filter in the first graphic and the error between them in the second graphic.

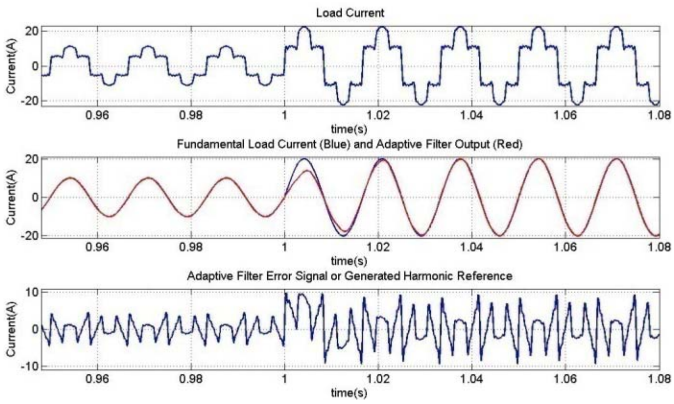

Fig. 5. Simulations results: 100% load current increase with adaptive FIR filter.

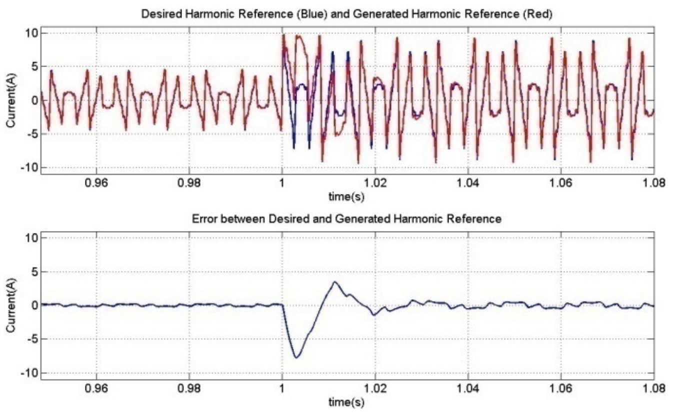

Fig. 6. Error between desired harmonic reference and generated harmonic reference by adaptive FIR filter with load current transient.

The transient behavior of the adaptive notch filter strategy, for a 100% load current variation, is presented in Fig. 7 and Fig. 8. The Fig. 7 shows the signals: the load current $d(n)$ in the first graphic, the fundamental of the load current (in blue) and the adaptive notch filter output $y(n)$ (in red) in the second graphic and the harmonic current reference $e(n)$ in the third graphic. The Fig. 8 shows the desired harmonic reference and the harmonic reference generated by the adaptive notch filter in the first graphic and the error between them in the second graphic.

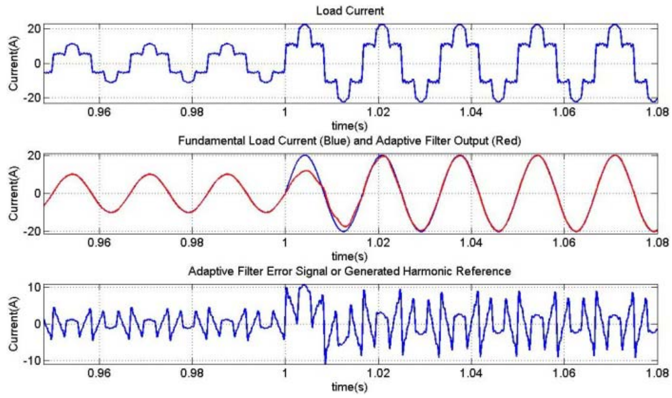

Fig. 7. Simulations results: 100% load current increase with adaptive notch filter.

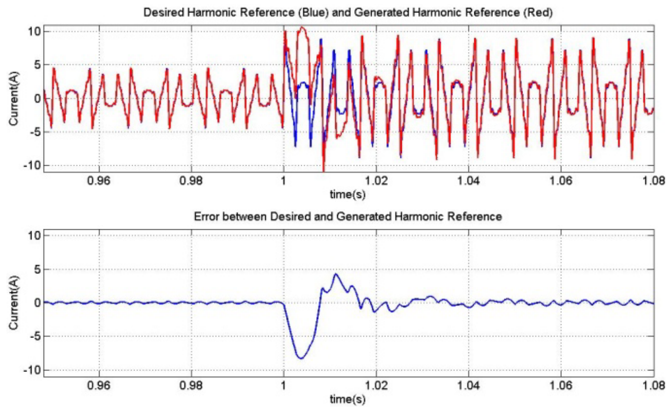

Fig. 8. Error between desired harmonic reference and generated harmonic reference by adaptive notch filter with load current transient.

The simulations demonstrate that there is a slightly difference between the steady-state error for the two strategies. The adaptive FIR filter presents an error a little bit bigger than for adaptive notch filter. The proposed algorithm for the step-size adjusts produces the same speed of convergence in both cases.

IV. EXPERIMENTAL RESULTS

The practical results are obtained from a shunt active power filter (SAPF) composed by three single-phase full H-bridges voltage source inverter (VSI) type, rated 35kVA, working in 40kHz switching frequency. The DC link voltage is adjusted in 500V. The smooth inductor used to connect the SAPF to power system is rated 5mH. The harmonic current injection is controlled by a VSI PWM. The power system voltage is 220V line to line. The load is a rated 100kVA, 6 pulse rectifier, CSI type. The shunt active power filter diagram is shown in the Fig. 9. In these experimental results a feedback configuration of SAPF was used, as illustrated in Fig. 8 three Hall current sensor measure the three phase source current. But the feedforward configuration, with the sensors measuring the three phase load current, can also be used. The control algorithm was implemented in a Texas DSP TMS320F2812.

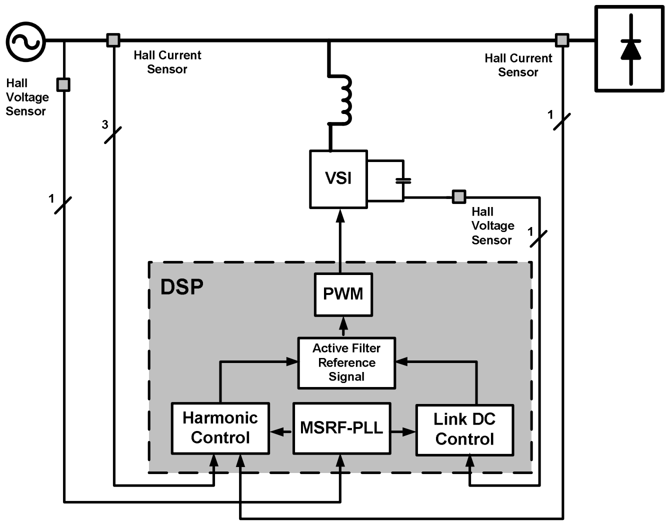

Fig. 9. Shunt Active Power Filter Block Diagram.

The Fig. 10 presents the load current (IL_A) of 27A, the compensation current (Ic_A) and the source current (Is_A) of phase A after compensation using an adaptive FIR filter strategy to extract the harmonic content. In Fig. 11 the harmonic spectrum of source current before and after compensation is shown, the current Total Harmonic Distortion (THDi) of Is_A is reduced from 28.9% to 7.56%.

The Fig. 12 presents IL_A, Ic_A and Is_A after compensation using an adaptive notch filter for a load current of 27A. The Fig. 13 shows the harmonic spectrum of source current, the THDi of Is_A is reduced from 28.9% to 8.29%.

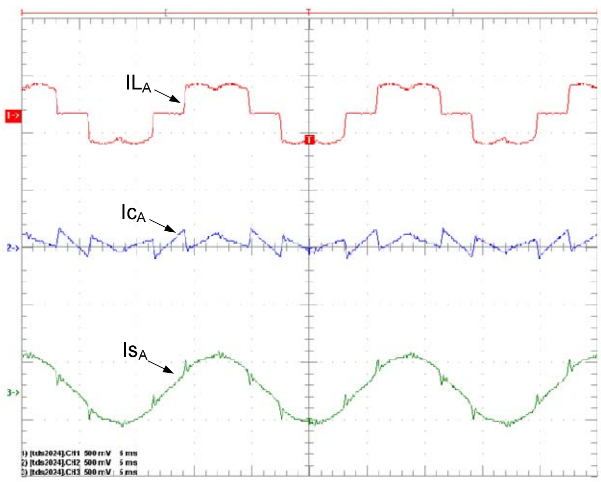

Fig. 10. Experimental results of adaptive FIR filter with a load current of 27A.

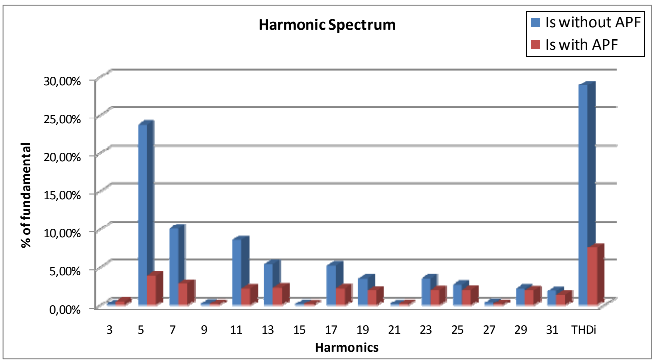

Fig. 11. Source current harmonic spectrum of the adaptive FIR filter.

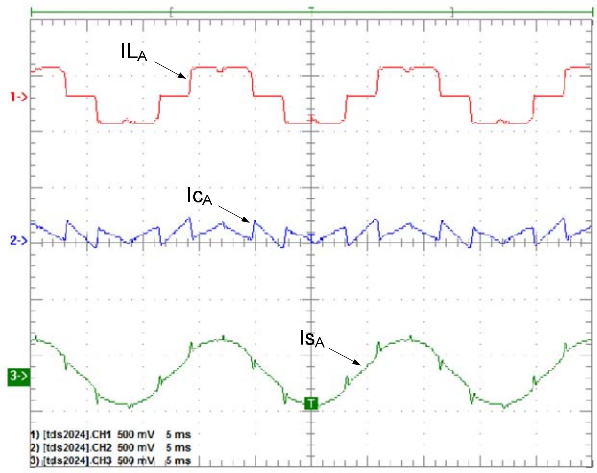

Fig. 12. Experimental results of adaptive notch filter with a load current of 27A.

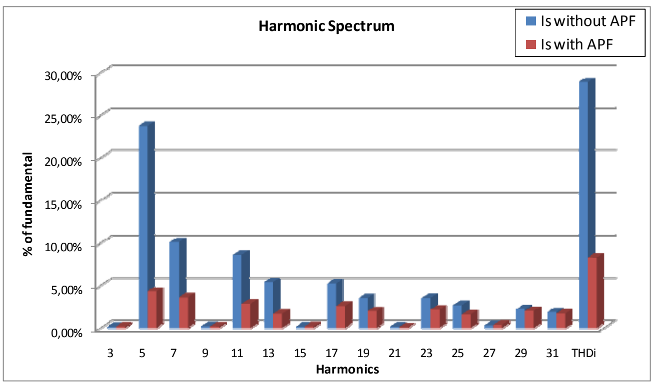

Fig. 13. Source current harmonic spectrum of the adaptive notch filter.

The Fig. 14 shows the adaptive notch filter transient behavior for a 100% load current variation. The result illustrates the convergence time Tc of 1.5 cycles. This result confirms that obtained by simulations.

The reduction of THDi in the two types of adaptive filter implemented has a small difference, the result for a FIR structure is a little better than adaptive notch filter. But the calculation time of the adaptive FIR filter and adaptive notch filter are completely different. The adaptive notch filter has just two coefficients to be adjusted, so its calculation time is significantly smaller than adaptive FIR filter that uses 32 coefficients for the same task.

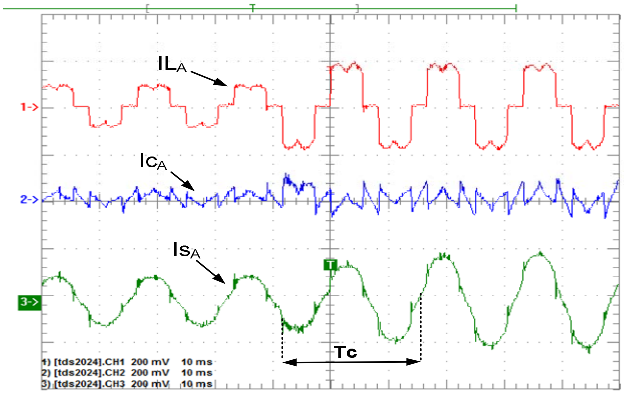

Fig. 14. Experimental results: 100% load current increase with adaptive notch filter.

V. CONCLUSION

In real time applications, as active power filters, the calculation time, the speed of convergence and steady-state error are critical matters to be taken in consideration during the design process. The proposed step-size adjustments algorithm, for the two types of adaptive filters, demonstrates good results concerning speed of convergence and steady-state error. The implementation with adaptive notch filter demonstrated to be the best choice related to calculation time, because adaptive notch filter has just two coefficients to be adjusted.

The results of the new strategy were faster than the results reported in the literature. The simulations and experimental results showed that the convergence time last 1.5 cycles of the fundamental, against the minimum of 2 cycles reported in the literature.

The adaptive notch filter algorithm has shown to be more interesting in terms of computational burden and steady-state error.

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